

Does Contract Commitment Reduce Customer Churn?

A causal analysis of the IBM Telco customer-churn dataset, produced end-to-end with the aitiolin Causality Agent.

Dataset: Churn.csv (IBM Telco; 21 raw variables, 7,043 customers)

Treatment: Contract (month-to-month → one-year → two-year, coded 0/1/2)

Outcome: Churn (customer left within the observation period, 0/1)

Confounder identified: tenure (months with the company)

Model source: two LLM-assisted agent drafts, statistically verified, merged by the analyst

Headline result

Moving a customer one contract tier up (e.g. from month-to-month to a one-year contract) **causally reduces churn probability by about 14.2%**, after adjusting for tenure - the classic confounder, since long-standing customers both choose longer contracts and churn less anyway. The naive, unadjusted comparison overstates the effect by ~58% (-0.224 instead of -0.142); even under a worst-case confounding scenario the effect stays meaningfully negative (-0.057). Unlike a spurious correlation, this effect **survived every refutation test** - contract commitment is a genuine, actionable lever against churn.

1. The question and the data

Telco customers on month-to-month contracts churn far more often than customers on one- or two-year contracts. But does the contract itself *cause* retention - or do loyal, long-tenure customers simply self-select into longer contracts? The distinction decides whether pushing contract upgrades (discounts for annual commitment, default-term changes) is worth real money. Tenure is the obvious confounder: it plausibly drives both contract choice and churn risk, inflating the raw correlation.

Data: the IBM Telco customer-churn dataset (7,043 customers, 21 columns) - demographics (gender, senior citizen, partner, dependents), subscribed services (phone, internet, security, backup, support, streaming), account attributes (tenure, contract, billing, payment method, charges) and the churn label.

Data preparation (Causality Agent, stages 1-2)

The agent's profiler flagged **customerID** as an identifier (7,043 unique values - no causal signal) and found the dataset's well-known quirk: **TotalCharges** is stored as text and contains 11 blank entries (customers with zero tenure). Both default-accepted steps were applied:

Column	Action	Result
customerID	drop column	Dropped column 'customerID'.
TotalCharges	impute missing	Imputed 11 value(s) in 'TotalCharges' with median 1397.

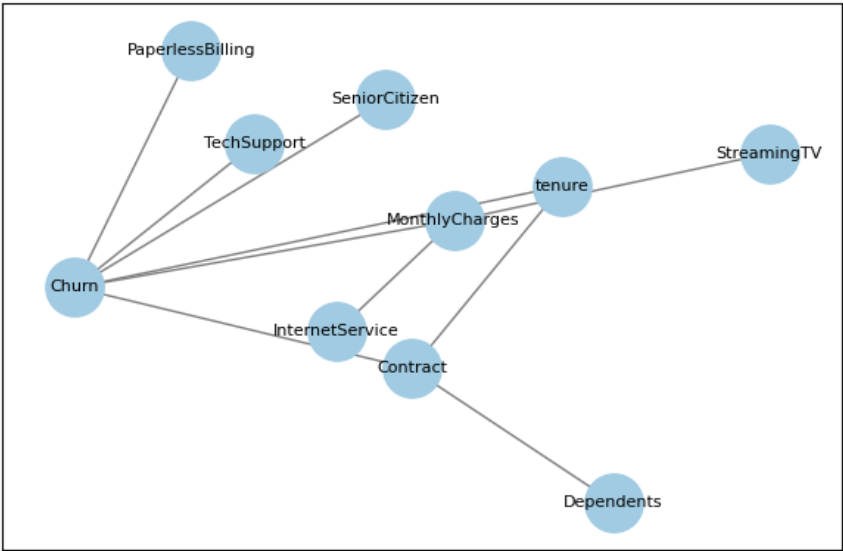
Result: 21 columns reduced to 20; all 7043 rows retained.

2. Building the causal model: two drafts and a merge

The agent's **first draft** proposed 12 edges - every one of them pointing directly into Churn. That “star” shape is exactly the structure a naive feature-importance view produces: it contains no confounding paths, so identification trivially requires no adjustment and the estimate equals the raw correlation. Re-prompting with explicit instructions to model relationships *among* the covariates produced a **second, structural draft** (tenure → Contract, Dependents → Contract, InternetService → MonthlyCharges, ...) - but that draft dropped the tenure → Churn edge the first draft had verified at full confidence. The analyst's merge step combined the statistically-supported edges of both drafts, which is precisely the human-in-the-loop review the app is designed around. The merged model:

Edge	Verifier verdict	Confidence
tenure -> Contract	supported	1.00
InternetService -> MonthlyCharges	supported	1.00
Contract -> Churn	supported	1.00
MonthlyCharges -> Churn	supported	0.83
SeniorCitizen -> Churn	supported	0.83
Dependents -> Contract	supported	1.00
TechSupport -> Churn	supported	1.00
StreamingTV -> MonthlyCharges	supported	1.00

Edge	Verifier verdict	Confidence
PaperlessBilling -> Churn	supported	0.83
tenure -> Churn	supported	1.00



Rendered causal graph as saved to the canvas.

Agent's reasoning (draft 2): This causal structure is plausible as it captures the relationships between contract types and churn, which is the primary outcome of interest. The treatment candidates, particularly the contract type, are actionable levers that the company can manipulate to potentially reduce churn rates. The edges illustrate how demographic factors and service choices influence both contract selection and churn, highlighting the complexity of customer retention. The biggest structural uncertainty lies in unobserved confounders such as customer satisfaction and service quality, which may significantly impact both contract choices and churn behavior.

3. Identification: tenure is the key

With tenure pointing into both Contract and Churn, the backdoor path **Contract** ← **tenure** → **Churn** must be blocked. Identification analysis confirms the effect is identifiable with the minimal adjustment set **{tenure}** (badge: trust, n = 7,043). Dependents also influences Contract but has no direct path to Churn, so it needs no adjustment; MonthlyCharges, TechSupport and the other direct causes of Churn sit off the backdoor path.

Check	Status	Detail
Graph is a DAG	pass	Identification assumes an acyclic causal graph.
Treatment variation	pass	Treatment must vary to support estimation.
Outcome variation	pass	Low outcome variation weakens the analysis.
Backdoor path review	warn	Open undirected paths may indicate unresolved confounding.

4. The estimate - naive vs. adjusted vs. worst case

Model	What it assumes	Effect per contract tier
Naive (no adjustment)	Naive view: only direct edges into Churn, no confounding structure - equivalent to the raw comparison.	-0.2237
Adjusted model (headline)	The merged model: adjusts for tenure (the identified confounder).	-0.1418
Confounder-stressed	Worst case: every direct cause of Churn also assumed to drive Contract, forcing maximal adjustment.	-0.0574

The three numbers bracket the truth. The naive estimate (-0.224) overstates the effect by ~58% because part of the raw contract-churn gap is really tenure: loyal customers pick long contracts *and* stay. After blocking that path, one contract tier is still worth **14.2% lower churn probability** - and even under the deliberately hostile assumption that every churn driver also shapes contract choice, 5.7% of the effect remains. Direction is unanimous; only magnitude varies (74% spread, verdict: moderate).

5. Robustness and refutation checks

Baseline: backdoor.linear_regression, effect -0.1418.

Refutation tests - all four passed

Test	Result	Interpretation
placebo treatment	status: ok, delta: 1.42e-01	With a randomly permuted fake treatment the effect collapses to exactly 0 - the estimate is not an artifact of the pipeline.
dummy outcome	status: ok, delta: 1.42e-01	Against a synthetic outcome the effect vanishes (-0.0002) - the model does not manufacture effects.
random common cause	status: ok, delta: 4.56e-06	Adding an irrelevant random covariate moves the estimate by only 0.000005 - adjustment is stable.
data subsample	status: ok, delta: 4.51e-04	On an 80% random subsample the estimate shifts by just 0.0005 (-0.1413 vs -0.1418) - the effect does not depend on particular customers.

Confounder-strength sensitivity sweep

Assumed unobserved-confounder strength	Adjusted effect
0.1	-0.1368
0.2	-0.1319
0.3	-0.1269
0.4	-0.1219
0.5	-0.1170

Even granting a hypothetical unobserved confounder half as strong as the observed adjustment (strength 0.5), the effect only shrinks from -0.142 to -0.117. Contrast this with the World Bank health-spending analysis in the companion example, where the same sweep drove the effect essentially to zero: **this effect is robust where that one was not** - which is exactly the discrimination a causal toolchain should provide.

The formal partial-R² sensitivity statistic is unavailable for this graph: DoWhy's implementation does not support models containing effect modifiers (direct causes of the outcome that are not confounders - here

MonthlyCharges, TechSupport, SeniorCitizen, PaperlessBilling and tenure's direct edge). This is a documented library limitation, not a failed check; the manual sweep above covers the same question.

6. Conclusions

- **Contract commitment causally reduces churn.** One tier of additional commitment is worth roughly 14 percentage points of churn probability after adjusting for tenure - a large effect against a ~27% base churn rate.
- **But the naive number is inflated.** Reading the raw contract-churn gap (-0.224) as causal would overstate the payoff of contract-upgrade campaigns by more than half; budget decisions should use the adjusted figure.
- **The effect is robust.** All four refutation tests passed cleanly, the estimate is stable across subsamples, and even worst-case structural assumptions leave a -5.7 percentage-point effect.
- **Actionable read:** incentives that move month-to-month customers onto annual contracts plausibly buy real retention - and because the effect survives adjustment for tenure, it is not just loyal customers self-selecting. A/B testing contract incentives on new cohorts would be the natural confirmatory next step.

Limitations

- **Ordinal treatment coding:** Contract is coded 0/1/2, so the estimate assumes the month-to-month→one-year step equals the one-year→two-year step. Estimating the two steps separately (e.g. binary contrasts) would refine this.
- **Linear probability model:** the binary churn outcome is estimated by linear regression; effects near probability boundaries can be distorted relative to a logistic specification.
- **Only tenure is adjusted for:** the identified adjustment set follows from the drawn graph. Unmeasured traits - price sensitivity, satisfaction, competitor offers - could still confound contract choice, though the sensitivity sweep bounds their plausible impact.
- **Cross-sectional snapshot:** a single observation window; no within-customer variation over time is exploited.

Reproducing this analysis

Upload Churn.csv (repository root) in the app, run the Causality Agent stages 1-3 (accept the two cleaning steps), re-prompt the model suggestion with a request for confounding structure among covariates, merge/lock the supported edges on the canvas (ensuring tenure points into both Contract and Churn), select Contract as treatment and Churn as outcome, then run estimation, the Robustness Dashboard, and the competing-model comparison from the agent panel.